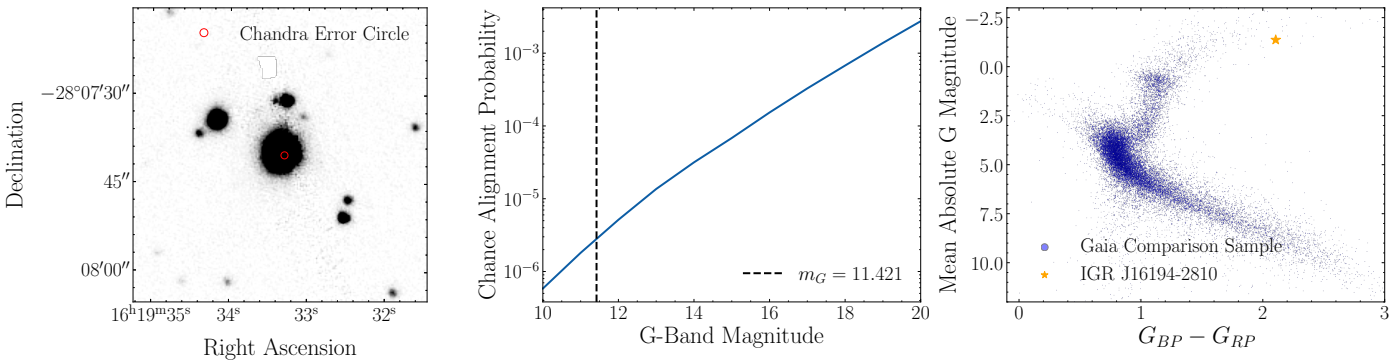

Figure 1. Left Panel: PanSTARRS field of view ($50'' \times 50''$) in g-band. The image has been rescaled to enhance contrast. The *Chandra* error circle of 0.6 arcseconds around the X-ray localization is shown. Middle Panel: Chance alignment probability of optical counterparts in the field of IGR J16194-2810 with the X-ray source detected by *Chandra* as a function of G-band apparent magnitude. At the apparent magnitude of the giant, the chance coincidence probability of $\sim 3 \times 10^{-6}$ is negligible. Right Panel: Location of IGR J16194-2810 on a extinction-corrected *Gaia* CMD. A sample of stars within 1° of the target with well-constrained parallaxes is shown for comparison. The companion star in IGR J16194-2810 has evolved far up the red giant branch.

of the system is $\approx$ 192 days. The normalized and phased light curves (with outliers removed) are plotted for both filters in the bottom panels of Figure 2. These light curves are compared to the best-fit ellipsoidal variability models derived from our joint fitting of all available data (see Section 5.1).

We retrieved apparent magnitudes in other optical bands from the APASS catalog (Henden et al. 2016) and in the infrared from the 2MASS survey (Skrutskie et al. 2006) and *WISE* mission (Wright et al. 2010). The source is saturated in deeper surveys such as PanSTARRS (Chambers et al. 2016) and is outside the GALEX footprint (Martin et al. 2005). These apparent magnitudes are converted to observed fluxes and are plotted against the central wavelength of their corresponding filters in Figure 3. The vertical error bars are derived based on the errors in the apparent magnitudes, while the horizontal error bars represent the effective widths of the corresponding bands. The photometry is compared to the best-fit model SED derived from our joint fitting of all available data (see Section 5.3).

4.2. *Spectroscopy*

4.2.1. *FEROS*

We observed IGR J16194-2810 17 times with the Fiberfed Extended Range Optical Spectrograph (FEROS; Kaufer et al. 1999) on the 2.2m ESO/MPG telescope at La Silla Observatory (programs P111.A-9003, P112.A-6010, P113.26XB). We use exposure times of 900–1200s. The resulting spectra have resolution $R \approx 50,000$ over a spectral range of 350–920 nm. The typical signal-to-noise ratio (SNR) per pixel at 5800 Å is ≈ 15.

We reduced the data using the CERES pipeline (Brahm et al. 2017), which performs bias-subtraction, flat fielding, wavelength calibration, and optimal extraction. The pipeline measures and corrects for small shifts in the wavelength solution over the course of a night using simultaneous observations of a ThAr lamp with a second fiber.

We calculate RVs by cross-correlating a synthetic template spectrum with each order individually and then report the mean RV across 15 orders with wavelengths between 4500 and 6700 Å. We calculate the uncertainty on this mean RV from the dispersion between orders; i.e., $\sigma_{\rm RV} \approx {\rm std\,(RVs)}/\sqrt{15}$. We used a Kurucz model spectrum (Kurucz 1979, 1993) with $T_{\rm eff} = 3750$ K, $\log g = 1.5$, and $[{\rm Fe/H}] = 0$ from the `BOSZ` library (Bohlin et al. 2017) as a template. Adopting a linear limb darkening coefficient based on Claret & Bloemen (2011), we use the direct integration technique implemented by Carvalho & Johns-Krull (2023) to apply rotational broadening to the template spectrum assuming a projected equatorial rotational velocity of $v \sin i = 15$ km s^{-1} (see Section 5.5).

All of our measured RVs (with heliocentric corrections applied) are presented in Table 3. The median uncertainty of the FEROS RVs is 0.22 km s^{-1}.

4.2.2. *MIKE*

We observed IGR J16194-2810 two times with the Magellan Inamori Kyocera Echelle (MIKE) spectrograph on the Magellan Clay telescope at Las Campanas Observatory (Bernstein et al. 2003). In one observation, we used the 0.7" slit with an exposure time of 840s, yielding spectral resolution $R \approx 32,500$ over a spectral range of 333–968 nm. In this spectrum, the typical SNR per pixel at 5800 Å is ≈ 70. In another observation, we use the 0.35" slit with an exposure time of 1800s, yielding higher resolution by a factor of 2, but a lower SNR per pixel at 5800 Å of ≈ 55. We reduced the spectra with the MIKE pipeline within CarPy (Kelson et al. 2000; Kelson 2003). To combine the orders, we perform an inverse variance-weighted average. We normalize the resulting spectrum using a polynomial spline fit to continuum wavelengths.

5. ANALYSIS

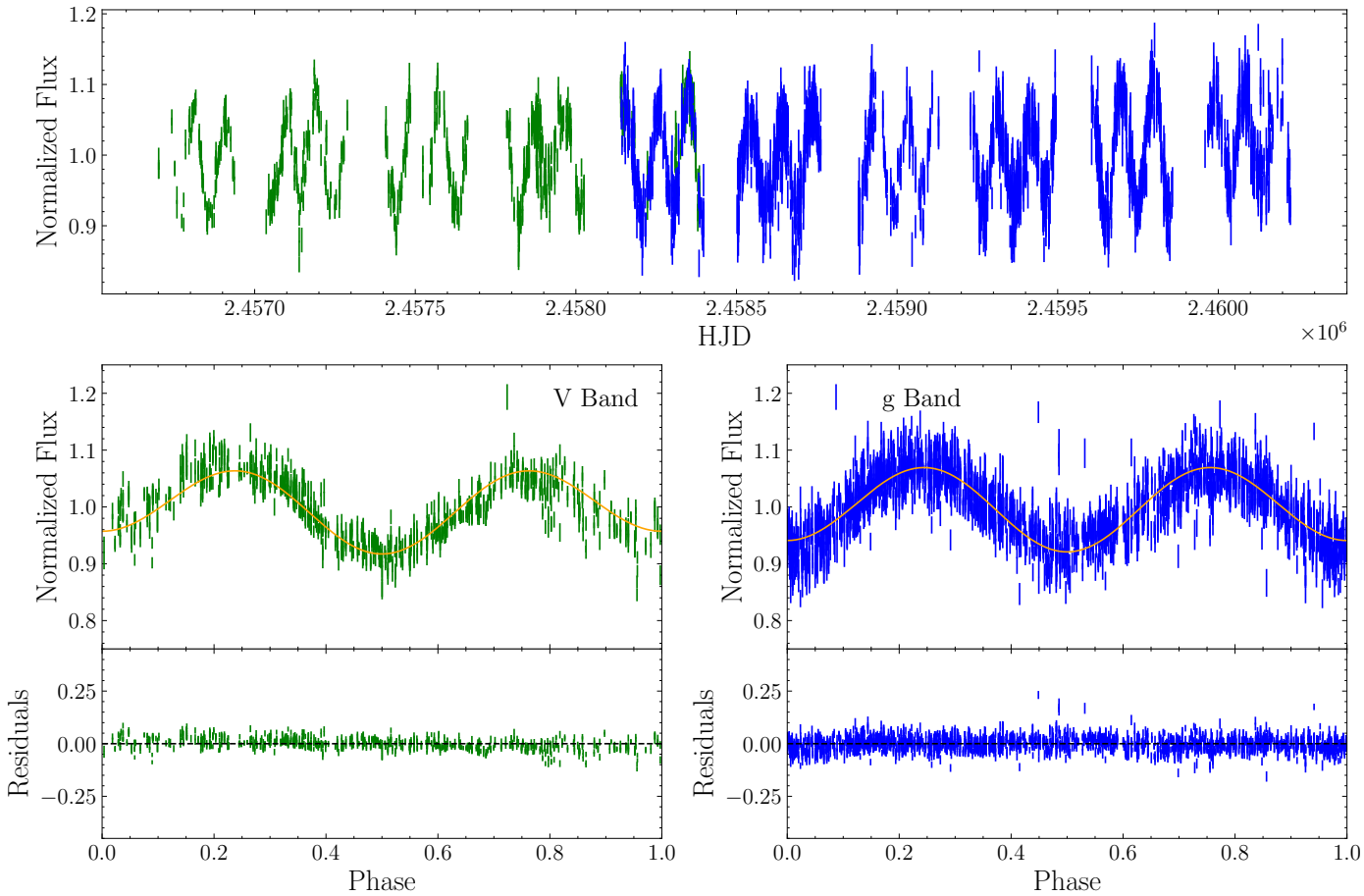

Figure 2. In the top panel, we show the V-band (green) and g-band (blue) light curves of IGR J16194-2810 obtained from the ASAS-SN survey. In the bottom panels, those light curves are phased using the best-fit orbital period of 192.73 days. The orange curves represent the best-fit ellipsoidal light curve model in each filter. In both filters, the corresponding residuals are consistent with zero, implying that the ellipsoidal light curve model explains almost all of the observed photometric variability.

5.1. *Light Curve Modeling*

Archival ASAS-SN light curves for the red giant companion in the V and g bands from the ASAS-SN survey are provided in the top panel of Figure 2. The normalized and phase-folded light curves, displaying approximately ellipsoidal variability, are provided in the bottom panels of Figure 2. We describe the ellipsoidal variation of the flux of the red giant in a given filter with a two-harmonic model:

$$F = \bar{F} + \sum_{i=1}^{2} A_i \cos\left(\frac{2\pi i(t - T_0)}{P}\right) \quad (1)$$

Here, P is the orbital period, T_0 is the epoch of conjunction, $\bar{F}$ is the mean flux in the given filter, and A_i is the variability amplitude associated with the ith harmonic. We expect A_2 to be the dominant component (Morris & Naftilan 1993; Gomel et al. 2021). We predict A_2 for a given choice of mass ratio, Roche lobe filling factor, inclination, and limb- and gravity-darkening coefficients using the prescription of Morris & Naftilan (1993) (which assumes circular orbits, tidal locking, and linear limb-darkening and gravity-darkening laws) and apply the correction factor of Gomel et al. (2021) (which is relevant at large Roche lobe filling factors). Unlike Hinkle et al. (2024), we do not assume that the red giant fills its Roche lobe.

We initially leave $\bar{F}$ and A_1 as free parameters, and fit the ASAS-SN light curves separately. We then fix $\bar{F}$ and A_1 to the best-fit values derived from this initial fit when performing a joint fit to all available data. We also fix the linear limb-darkening and gravity-darkening coefficients in each filter based on Claret & Bloemen (2011). When predicting A_2, the Roche lobe-filling factor is derived from the NS mass, red giant mass, red giant radius, and orbital period. Here, the red giant radius is predicted from the red giant's age, initial mass, and initial metallicity (see Section 5.6), while the NS mass, red giant mass, and orbital period (along with the orbital inclination) are left as free parameters.

We plot the dependence of the ellipsoidal variability amplitude A_2 on the mass of the red giant in both filters (which have different limb-darkening and gravity-darkening coefficients) in Figure 4. At larger red giant masses, the primary-to-secondary mass ratio and red giant Roche lobe filling fac-

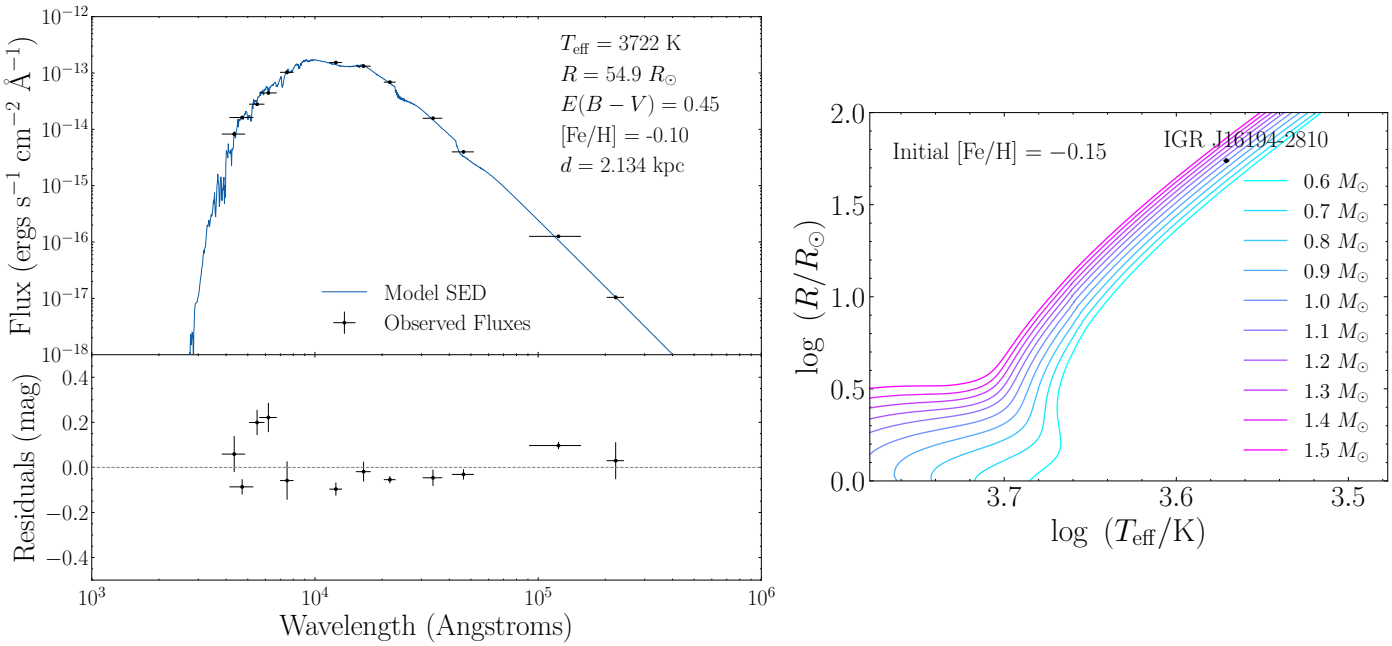

Figure 3. Left Panel: Spectral energy density model and associated residuals for the red giant star in IGR J16194-2810. A model SED is plotted based on the MAP parameters of the giant (see Table 4). The radius and effective temperature inferred from the SED analysis allow us to place tight constraints on the giant's mass. Right Panel: Comparison of the best-fit radius and effective temperature of the red giant to MIST single-star evolutionary models. Assuming an initial metallicity of [Fe/H] $= -0.15$, the location of the red giant on the diagram implies an initial luminous star mass of $\approx 1.0\,M_\odot$.

tor both decrease, causing A_2 to decrease as well. Furthermore, at more face-on inclinations, the variation in the geometric cross-section of the red giant is not as evident, which also causes A_2 to decrease. As a result, the respective constraints on A_2 from the observed light curve and the red giant mass from the observed SED imply that the orbital inclination of the system must be greater than 75°.

We simultaneously fit the ellipsoidal variation in both ASAS-SN filters, assuming the same underlying physical parameters. The log-likelihood term for the light curve data assumes Gaussian uncertainties:

$$\ln L_{\rm LC} = -\frac{1}{2}\sum_t \left(\frac{F_{{\rm obs},t} - F_{{\rm pred},t}}{\sigma_{F,t}}\right)^2 \qquad (2)$$

The sum is taken over all observation times and all filters. The best-fit light curve models and their corresponding residuals (taken from the joint fit described in Section 5.6) are plotted in the bottom panels of Figure 2. In both filters, the residuals are consistent with zero, implying that ellipsoidal modulation can explain almost all of the observed photometric variability.

5.2. *Fitting the Radial Velocity Curve*

We measured radial velocities (RVs) as described in Section 4. We plot the resulting RVs in Figure 5. We fit the RVs with a Keplerian two-body model:

$$\mathrm{RV}(t) = \gamma - K\sin\left(\frac{2\pi(t-T_0)}{P}\right) \qquad (3)$$

Here, γ is the center-of-mass RV, K is the RV semi-amplitude, and P and T_0 are the orbital period and epoch of conjunction, as before. We assume that the orbit is tidally circularized and synchronized, which is justified for binaries displaying ellipsoidal variability (Zahn 1977; Lurie et al. 2017). The log-likelihood term for the RV data assumes Gaussian uncertainties:

$$\ln L_{\rm RV} = -\frac{1}{2}\sum_t \left(\frac{\mathrm{RV}_{{\rm obs},t} - \mathrm{RV}_{{\rm pred},t}}{\sigma_{{\rm RV},t}}\right)^2 \qquad (4)$$

We present the best-fit RV curve (taken from the joint fit described in Section 5.6), along with its residuals, in the left panel of Figure 5. Assuming an edge-on orbit, the observed RV semi-amplitude of 26.3 ± 0.1 km s^{-1} implies a minimum companion mass of $1.21 \pm 0.02\,M_\odot$. We plot the inferred companion mass as a function of orbital inclination in the right panel of Figure 5. If we adopt the inclination constraint implied by the variability amplitudes of the ASAS-SN light curves (see Section 5.6), then we infer a companion mass of $1.23^{+0.05}_{-0.03}\,M_\odot$.

5.3. *Spectral Energy Distribution*

We use `pystellibs`[1] with the BaSeL library (Lejeune et al. 1997, 1998) to generate model SEDs, with the initial mass, initial metallicity, age, and parallax as free parameters. We assume a Cardelli et al. (1989) extinction law with

[1] https://mfouesneau.github.io/pystellibs/